The Cosmological Lithium Problem and the CPTG Lithium Solution

Carter L Glass Jr

Abstract

The cosmological lithium problem is the longstanding selective discrepancy between standard big-bang nucleosynthesis and the low primordial mass-seven abundance inferred from old metal-poor stellar systems while deuterium and helium remain successful comparison channels. Curvature Polarization Transport Gravity, or CPTG, treats this as a surviving mass-seven transport problem inside the primordial reaction network. The CPTG construction fixes the live mass-seven transport optical depth to $\tau_7 = \ln \pi$, giving the isolated-channel survival factor $1/\pi$ without fitting a lithium depletion coefficient. The transport acts directly on the live ^{7}Li and ^{7}Be channels rather than dividing a completed abundance after network evolution.

The present analysis implements the complete transported CPTG BBN comparison coordinate prospectively inside PRIMAT v0.3.2's native large reaction network. Two gate-off calibration rows determine the PRIMAT radiation translation using only the reported $N_{\rm eff}$ response, giving $dN_{\rm eff}/d(\Delta N_{\rm eff}) = 0.9999999999999343$ and the frozen target input $\Delta N_{\rm eff}^{\rm CPTG} = -0.07539322509392317$ before abundance-target execution. At $(\Omega_b h^2)_{\rm BBN} = 0.021898765370$, the network-native target returns $N_{\rm eff} = 2.968584073463995$, leaving an implementation/mapping-closure residual of 4.88×10^{-15} against the frozen CPTG numerical target 2.968584073464. This residual verifies the prospectively frozen translation inside PRIMAT; it is not an independent PRIMAT derivation or measurement of $N_{\rm eff}^{\rm CPTG}$. The live mass-seven gate then gives Y_7 survival 0.318366498376680, a relative deviation of 0.0177852% from $1/\pi$, while the relative changes in D/H, Y_p, and ^{3}He/H are 3.91×10^{-9}, 3.86×10^{-12}, and 5.59×10^{-10}, respectively, and the radiation coordinate is unchanged. The PRIMAT-reported ^{7}Li/H result moves from $5.2735647904 \times 10^{-10}$ to $1.6789263563 \times 10^{-10}$, corresponding to $+0.915705\sigma$ under the declared lithium comparison anchor. All 23 preregistered scientific gates pass. The result is therefore a prospective PRIMAT-network-native validation of the locked CPTG $A = 7$ transport law at the transported CPTG BBN comparison coordinate, with PRIMAT's own BBN background retained for the network evolution.

1 The Cosmological Lithium Problem

Big-bang nucleosynthesis is a high-precision early-universe test because light-element yields respond directly to the baryon-to-photon ratio, the relativistic radiation content, the neutron-proton conversion history, and the thermonuclear reaction network. The lithium problem is selective rather than global: standard calculations can preserve successful deuterium and helium behavior while producing substantially more surviving mass seven than is inferred from old metal-poor stellar atmospheres [1, 2, 3, 6].

Most surviving primordial mass seven is carried through ^{7}Be during the nuclear epoch and later contributes to lithium after electron capture. The relevant primordial network coordinate is therefore

$$Y_7 = Y_{^7\mathrm{Li}} + Y_{^7\mathrm{Be}}. \tag{1}$$

The lithium comparison anchor used in this paper is

$$^7\mathrm{Li/H}_{\mathrm{obs}} = (1.45 \pm 0.25) \times 10^{-10}. \tag{2}$$

The corresponding deuterium and helium comparison controls are

$$\mathrm{D/H}_{\mathrm{obs}} = (25.08 \pm 0.29) \times 10^{-6}, \tag{3}$$

and

$$Y_{p,\mathrm{obs}} = 0.245 \pm 0.003. \tag{4}$$

The deuterium, helium, and lithium values in Equations (2)–(4) are the declared observational comparison anchors used by the present analysis, consistent with the 2025 Review of Particle Physics BBN summary [5]. The stellar-lithium interpretation is treated more cautiously below.

A viable CPTG lithium solution must therefore preserve the successful light-element background, reduce the surviving mass-seven abundance, and do so without introducing a fitted lithium depletion factor.

1.1 Observational-anchor audit

The stellar lithium literature does not define one uniquely Gaussian primordial-lithium measurement. High-resolution metal-poor-star analyses have reported a plateau or plateau-like abundance substantially below standard BBN expectations, while also identifying temperature-scale, metallicity, and very-low-metallicity systematics. Asplund et al. estimated ^{7}Li/H in the approximate range 1.1–1.5×10^{-10} for their metal-poor sample [2]. Bonifacio et al. found a mean plateau level $A(\mathrm{Li}) = 2.10 \pm 0.09$ for their selected extremely metal-poor turnoff stars and emphasized the importance of temperature-scale systematics [3]. A later analysis of roughly two hundred metal-poor dwarfs and subgiants showed that the notion of a single universal plateau becomes increasingly problematic at the lowest metallicities [4].

For that reason, Equation (2) is treated as a *declared comparison anchor*, not as a claim that the full stellar literature is exactly represented by one Gaussian datum. It is used only to report a downstream observational pull. It is not used to choose the transport optical depth, gate window, native solver, baryon coordinate, nuclear rates, or scientific PASS/FAIL decision.

2 Relation to the Universal CPTG Nuclear-Reaction Framework

The locked mass-seven transport law originates in the fixed CPTG geometric pi-branch comparison construction [9]. In that parent authority, the identity is stated explicitly in the subsection *Primordial Nuclear and Mass-Seven Abundance-Transport Coordinates*: the geometric branch is fixed before downstream nuclear-reaction and lithium validation, and the integrated mass-seven transport depth is $\tau_7 = \ln \pi$, giving the isolated-channel survival factor $e^{-\tau_7} = 1/\pi$. The present paper does not derive or fit that value from lithium abundances; it applies and prospectively tests the already-fixed pi-branch law inside a native primordial reaction-network solver. The same value is carried into the campaign preregistration identified cryptographically in Section 9, before target-row execution and before the observational lithium diagnostic can be consulted. Here and below, *network-native* means that PRIMAT itself evolves the coupled abundance equations, including the live CPTG sink terms; it does not mean that PRIMAT's cosmological background has been replaced by the native CPTG Pi-branch expansion history.

The broader CPTG nuclear-reaction program supplies the reaction-space interpretation of that transport law. The universal parent theory organizes the validated primordial source geometry into four coupled sectors: the free-neutron/proton vertex, the deuterium bridge, the tritium/helium-3 closure sector, and helium-4 saturation [12]. Those sectors define the fixed source geometry, baryon and electric-charge constraints, three ordered transport directions, and one charge-constrained polarization direction of the rank-four light-sector source space.

Mass seven is not introduced as a fifth source sector in that foundation. Instead, it is a distinct live abundance-transport sector downstream of the four-sector source architecture. The live $A = 7$ term is therefore an abundance-sector realization of CPTG curvature transport, not a fifth coordinate of the baryon-conserving four-sector reaction simplex and not an asserted conventional nuclear-reaction channel. The commissioned CPTG nuclear-reaction paper subsequently carried the same already-locked mass-seven law through a source-to-reaction-to-abundance construction, providing an earlier independent implementation of the transport sector [11].

The computational companion independently places mass seven inside the PRIMAT-network-native domain of the complete processed chain and keeps it distinct from the non-native reduced-graph continuation beginning above PRIMAT's frozen native boundary [13]. The present paper is the dedicated complete-coordinate application, network-native implementation, and prospective validation record for that already-defined $A = 7$ transport law.

CPTG is organized as a geometric theory in which baryonic matter sources nonlinear curvature polarization and curvature transport. Its scalar laws are reductions of a native nonlinear geometric branch, not independent fitting rules imposed separately in each observational sector [8, 9].

The schematic CPTG action is

$$S_{\text{CPTG}} = \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R + \mathcal{L}_{\text{pol}} + \mathcal{L}_{\text{trans}} + \mathcal{L}_m\right]. \tag{5}$$

Variation with respect to the metric gives the schematic field equation

$$G_{\mu\nu} + \Pi^{\text{pol}}_{\mu\nu} + \Theta^{\text{trans}}_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}. \tag{6}$$

For the abundance problem, the operational ordering is

$$\begin{gathered}\text{CPTG-native geometric quantity}\\ \downarrow\\ \text{transported BBN abundance coordinate}\\ \downarrow\\ \text{locked live mass-seven transport law}\\ \downarrow\\ \text{network-native dynamic response}\\ \downarrow\\ \text{observational comparison.}\end{gathered} \tag{7}$$

This ordering is essential. The lithium result is not defined by a post-processing rescaling of a completed abundance.

3 BBN Coordinate and Light-Element Conversion

The CPTG pi branch separates the acoustic/CMB baryon coordinate from the nuclear reaction-network coordinate. The native curvature exponent is

$$p_C = \pi. \tag{8}$$

The acoustic exponent is

$$p_{\text{ac}} = 3 - \frac{\pi}{100} = 2.968584073464. \tag{9}$$

The acoustic transport factor is

$$\mathcal{G}_T = \frac{p_{\text{ac}}}{p_C} = 0.9449296585513721. \tag{10}$$

The acoustic baryon-to-photon coordinate is

$$\eta_{10}^{\rm ac} = 6.170380167710. \tag{11}$$

The BBN abundance coordinate is obtained by the transport-scaled conversion

$$\eta_{10}^{\rm BBN} = \eta_{10}^{\rm ac}\sqrt{\mathcal{G}_T}, \tag{12}$$

with

$$\eta_{10}^{\rm BBN} = 5.998071834744. \tag{13}$$

The values displayed in Equations (12) and above are rounded publication values, whereas the locked BBN target is evaluated from the full stored precision of the parent transport quantities. Recomputing from only the displayed values $\eta_{10}^{\rm ac} = 6.170380167710$ and $\mathcal{G}_T = 0.9449296585513721$ gives 5.998071834741609; the 2.39×10^{-12} difference from the locked value is therefore display-rounding provenance, not a distinct physical or fitted correction.

The corresponding physical baryon-density coordinate is

$$(\Omega_b h^2)_{\rm BBN} = 0.021898765370. \tag{14}$$

The CPTG comparison layer also identifies

$$N_{\rm eff}^{\rm CPTG} = p_{\rm ac} = 2.968584073464. \tag{15}$$

The displayed twelve-decimal value is the locked numerical comparison target used by the prospective PRIMAT radiation mapping.

The present PRIMAT experiment tests the transported CPTG BBN comparison tuple

$$\Theta_{\rm BBN}^{\rm CPTG} = \left(\eta_{10}^{\rm BBN}, N_{\rm eff}^{\rm CPTG}, \tau_7\right). \tag{16}$$

In this paper, the terms *complete transported coordinate* and *full-coordinate test* refer to this jointly specified BBN comparison tuple: transported baryon coordinate, transported radiation coordinate, and independently locked live mass-seven optical depth. PRIMAT retains its own native background and temperature–time evolution for the network calculation.

The parent CPTG comparison-coordinate construction previously reported the transported-coordinate reference values

$$Y_{p,\rm ref} = 0.245694338195, \tag{17}$$

and

$$10^6({\rm D/H})_{\rm ref} = 25.074383114428. \tag{18}$$

These are parent comparison-layer reference values, not substitutions for the PRIMAT network outputs and not acceptance gates in the prospective experiment below.

Equation (15) is implemented natively in the prospective PRIMAT test below. The PRIMAT radiation translation is determined before target abundance exposure from two gate-off calibration rows, one at $\Delta N_{\rm eff} = 0$ and one at $\Delta N_{\rm eff} = 0.01$. Only the reported native $N_{\rm eff}$ values enter the mapping. With

$$N_{\rm eff}(0) = 3.0439772985579183, \tag{19}$$

$$N_{\rm eff}(0.01) = 3.0539772985579177, \tag{20}$$

the empirical slope is

$$\frac{dN_{\rm eff}}{d(\Delta N_{\rm eff})} = 0.9999999999999343. \tag{21}$$

The resulting PRIMAT input is frozen before either target row is executed,

$$\Delta N_{\rm eff}^{\rm CPTG} = -0.07539322509392317. \tag{22}$$

Together with Equation (14), this defines the transported CPTG comparison coordinate supplied to the network-native PRIMAT validation.

4 Locked Mass-Seven Transport Law

The CPTG lithium mechanism is the fixed mass-seven optical-depth transport law inherited from the geometric pi-branch comparison construction [9]. The parent authority states $\tau_7 = \ln \pi$ in its subsection *Primordial Nuclear and Mass-Seven Abundance-Transport Coordinates* and explicitly places that lock before downstream nuclear-reaction and lithium validation. The law is therefore fixed before the present lithium calculation: the present work neither derives τ_7 from the observed lithium abundance nor fits its value to the PRIMAT endpoint. The same locked law was previously carried through the commissioned CPTG nuclear source-to-reaction-to-abundance construction [11]. For this PRIMAT campaign, the inherited optical depth and temporal embedding were then frozen in the preregistration whose SHA-256 identity is preserved in Section 9; target-row execution could not revise them. Consistent with the universal four-sector framework, this live term is a separate abundance-transport action rather than an additional baryon-simplex reaction coordinate.

For the actual network experiment, let $t_{\rm net}$ denote PRIMAT's BBN network-evolution time. The mass-seven gate is localized by the preregistered temperature window, so its implementation is tied directly to the network's native temperature–time evolution. For an isolated survival channel,

$$Y_{7,\rm final} = Y_{7,\rm raw} \exp\left[-\int \Gamma_{7,\rm trans}(t_{\rm net})dt_{\rm net}\right]. \tag{23}$$

The locked CPTG geometric condition is

$$\int \Gamma_{7,\rm trans}(t_{\rm net})dt_{\rm net} = \ln \pi. \tag{24}$$

Therefore,

$$Y_{7,\mathrm{CPTG}} = Y_{7,\mathrm{raw}} e^{-\ln \pi} = \frac{Y_{7,\mathrm{raw}}}{\pi}, \tag{25}$$

with locked isolated-channel survival factor

$$\frac{1}{\pi} = 0.3183098861837907. \tag{26}$$

The appearance of $1/\pi$ here belongs specifically to the independently locked mass-seven abundance-transport identity $e^{-\tau_7} = 1/\pi$. It is distinct from the use of $1/\pi$ as a diagnostic synchronization-rate benchmark in the separate CPTG cosmological-horizon analysis [10]; no horizon synchronization rate is used to normalize the lithium gate.

At source-network level, the live lithium and beryllium channels are evolved with

$$\frac{dY_{^7\mathrm{Li}}}{dt_{\mathrm{net}}} \longrightarrow \left(\frac{dY_{^7\mathrm{Li}}}{dt_{\mathrm{net}}}\right)_{\mathrm{BBN}} - \Gamma_{7,\mathrm{trans}}(t_{\mathrm{net}}) Y_{^7\mathrm{Li}}, \tag{27}$$

$$\frac{dY_{^7\mathrm{Be}}}{dt_{\mathrm{net}}} \longrightarrow \left(\frac{dY_{^7\mathrm{Be}}}{dt_{\mathrm{net}}}\right)_{\mathrm{BBN}} - \Gamma_{7,\mathrm{trans}}(t_{\mathrm{net}}) Y_{^7\mathrm{Be}}. \tag{28}$$

For the prospective PRIMAT implementation, the temporal embedding is frozen as a log-cosine window from 0.0175 to 0.0040 MeV. Its normalization is fixed by Equation (24). Neither the window nor the integrated optical depth is selected from a lithium endpoint.

4.1 Exact optical-depth law versus live-network realization

Equations (25)–(26) are exact for the isolated survival channel. A native BBN network is not isolated: ordinary production, destruction, and redistribution continue while the transport term is active. The realized gate-off/gate-on endpoint ratio is therefore a network-response object and need not be algebraically identical to $1/\pi$, even though the inserted optical depth remains exactly normalized to $\ln \pi$.

This distinction turns the small native deviation from $1/\pi$ into a quantitative test. The prospective campaign freezes a tolerance on that network-response deviation before gated endpoints are opened. Agreement is then measured rather than imposed.

5 Prospective PRIMAT-Network-Native Complete-Coordinate Validation

5.1 Authority and execution design

PRIMAT v0.3.2 is the primary numerical reference implementation for the present test [6, 7]. The validation uses a copied PRIMAT source tree so the installed authority tree remains unchanged. The live calculation forces the Python backend, uses the native large network, numerical precision 10^{-10}, a 4000-point nuclear-rate grid, neutron lifetime 878.4 s, and the radiative, finite-mass, thermal, spectral-distortion, weak-rate-normalization, plasma-QED, and nuclear-QED corrections fixed by the preregistration.

The direct sink species are exactly ^{7}Li and ^{7}Be. Matching diagonal sink terms are inserted into PRIMAT's analytic Jacobian. The complete prospective execution consists of four native large-network solves:

1. a gate-off calibration row at $\Delta N_{\mathrm{eff}} = 0$;
2. a gate-off calibration row at $\Delta N_{\mathrm{eff}} = 0.01$;
3. a full-coordinate target row with the mass-seven gate off;
4. the same full-coordinate target row with the locked mass-seven gate on.

The first two rows may determine only the radiation-coordinate mapping in Equations (21)–(22). No abundance endpoint, including lithium, may enter that mapping. The mapping is hash-frozen before the two target rows execute, and the target rows are forbidden from modifying it.

5.2 Frozen acceptance conditions

The scientific gates are fixed before target-row abundance exposure. They require:

- the calibration slope to differ from unity by no more than 10^{-8};
- the radiation mapping to be frozen before target execution and to contain no abundance feedback;
- the gate-off and gate-on target values of N_{eff} each to agree with Equation (15) within 5×10^{-9} absolute;
- the gate-on/gate-off relative change in N_{eff} to be no larger than 10^{-10};
- the internal Y_7 survival ratio and PRIMAT's reported mass-seven/H survival ratio each to agree with $1/\pi$ within 0.5% relative;
- gated-versus-baseline relative changes in D/H, Y_p, and ^{3}He/H no larger than 10^{-6};
- native gate-off D/H and Y_p to lie within 2σ of the declared observational controls;
- an independent-grid optical-depth reconstruction within 2×10^{-6} relative of $\ln \pi$, with the direct quadrature reconstruction satisfying its frozen normalization test;
- nonzero runtime exercise of both the RHS and analytic-Jacobian sink paths;
- finite and nonnegative native final states; and
- unchanged installed PRIMAT authority source and verified import of the copied patched source.

The observational lithium comparison is report-only. It cannot alter the radiation mapping, transport normalization, temporal window, baryon coordinate, nuclear rates, solver settings, or pass/fail gates.

5.3 Runtime and normalization controls

The four native large-network solves complete in 331.76 s of recorded solver runtime. In the gated target row, runtime instrumentation records 15,562 nonzero RHS gate calls and 440 nonzero analytic-Jacobian gate calls. The direct quadrature reconstructs $\ln \pi$ with zero reported relative discrepancy at stored precision, while an independent sampled-grid reconstruction gives relative error

$$1.49682 \times 10^{-7}, \tag{29}$$

comfortably inside the frozen 2×10^{-6} requirement. The production transcript preserves a roundoff-limited adaptive-quadrature warning associated with the auxiliary gate-weight integration; it does not abort the native solve, and both the independently sampled normalization check and the live-network survival gate pass.

6 PRIMAT Network-Native Complete-Coordinate Result

6.1 Radiation-coordinate closure

The native two-row calibration gives

$$\frac{dN_{\rm eff}}{d(\Delta N_{\rm eff})} = 0.9999999999999343, \tag{30}$$

$$\left| \frac{dN_{\rm eff}}{d(\Delta N_{\rm eff})} - 1 \right| = 6.57 \times 10^{-14}. \tag{31}$$

The frozen translation in Equation (22) then produces

$$N_{\rm eff,baseline} = 2.968584073463995, \tag{32}$$

$$N_{\rm eff,gated} = 2.968584073463995. \tag{33}$$

The absolute discrepancy from the frozen numerical CPTG target in Equation (15) is

$$\left| N_{\rm eff,PRIMAT} - N_{\rm eff}^{\rm CPTG} \right| = 4.88 \times 10^{-15}, \tag{34}$$

and the gate-induced change in $N_{\rm eff}$ is zero at the reported precision. The 4.88×10^{-15} residual is an implementation/mapping-closure metric after the abundance-independent translation has been frozen; it is not an independent PRIMAT prediction of $N_{\rm eff}^{\rm CPTG} = p_{\rm ac}$, nor is any deuterium, helium, or lithium abundance used to construct that radiation mapping.

6.2 Native background and live mass-seven response

At the complete transported CPTG baryon-plus-radiation comparison coordinate, the gate-off PRIMAT network background is

$$\mathrm{D/H}_{\text{baseline}} = 2.5066925358359854 \times 10^{-5}, \tag{35}$$

$$Y^{\text{BBN}}_{p,\text{baseline}} = 0.24576024693340703, \tag{36}$$

$$(^3\mathrm{He/H})_{\text{baseline}} = 1.0503519370586517 \times 10^{-5}, \tag{37}$$

$$(^7\mathrm{Li/H})_{\text{baseline}} = 5.273564790371440 \times 10^{-10}, \tag{38}$$

$$Y_{7,\text{baseline}} = 3.977207473843914 \times 10^{-10}. \tag{39}$$

Relative to Equations (3) and (4), the native gate-off background lies at -0.045085σ in D/H and $+0.253416\sigma$ in Y_p, respectively.

An informative internal comparison is provided by the preregistered $\Delta N_{\text{eff}} = 0$ calibration row, which uses the same transported baryon coordinate and the same gate-off PRIMAT large-network configuration while retaining the native standard-radiation background. That row gives $N_{\text{eff}} = 3.0439772985579183$, $\mathrm{D/H} = 2.5323441770922762 \times 10^{-5}$, and $Y_p^{\text{BBN}} = 0.2467730506356945$, corresponding to residuals of $+0.839454\sigma$ and $+0.591017\sigma$ in the two declared background controls. At the independently fixed CPTG radiation coordinate, those residuals become -0.045085σ and $+0.253416\sigma$. If the two declared controls are summarized as independent Gaussian diagnostics, the corresponding two-observable statistic decreases from $\chi^2_{\mathrm{D,Y_p}} = 1.05398$ to 0.066252, an improvement by a factor of 15.91. This comparison is not a global cosmological likelihood analysis and does not by itself establish a universally preferred value of N_{eff}; within the PRIMAT-network-native BBN test at fixed transported baryon coordinate, it shows that the independently derived CPTG radiation coordinate simultaneously improves agreement with both preregistered background observables. No deuterium, helium, or lithium abundance enters the radiation-coordinate derivation.

With the frozen live mass-seven transport enabled, PRIMAT gives

$$\mathrm{D/H}_{\text{gated}} = 2.5066925456419876 \times 10^{-5}, \tag{40}$$

$$Y^{\text{BBN}}_{p,\text{gated}} = 0.24576024693245754, \tag{41}$$

$$(^3\mathrm{He/H})_{\text{gated}} = 1.0503519364716970 \times 10^{-5}, \tag{42}$$

$$(^7\mathrm{Li/H})_{\text{gated}} = 1.6789263562731326 \times 10^{-10}, \tag{43}$$

$$Y_{7,\text{gated}} = 1.2662096167652484 \times 10^{-10}. \tag{44}$$

The internal mass-seven survival factor is therefore

$$\frac{Y_{7,\text{gated}}}{Y_{7,\text{baseline}}} = 0.318366498376680, \tag{45}$$

with relative deviation

$$\left|\frac{(Y_{7,\mathrm{gated}}/Y_{7,\mathrm{baseline}}) - (1/\pi)}{1/\pi}\right| = 1.77852 \times 10^{-4} = 0.0177852\%. \tag{46}$$

PRIMAT's reported ^{7}Li/H survival factor is 0.318366498376685, giving the same relative agreement with $1/\pi$ to the displayed precision.

The corresponding non-mass-seven disturbances are

$$|\Delta(\mathrm{D/H})|_{\mathrm{rel}} = 3.91 \times 10^{-9}, \tag{47}$$

$$|\Delta Y_p|_{\mathrm{rel}} = 3.86 \times 10^{-12}, \tag{48}$$

$$|\Delta(^3\mathrm{He/H})|_{\mathrm{rel}} = 5.59 \times 10^{-10}, \tag{49}$$

$$|\Delta N_{\mathrm{eff}}|_{\mathrm{rel}} = 0. \tag{50}$$

Relative to the declared comparison anchor in Equation (2), the raw PRIMAT lithium result is $+15.2943\sigma$, while the gated result is

$$\Delta_{^7\mathrm{Li/H,gated}} = +0.915705\sigma. \tag{51}$$

Because lithium is excluded from every hard acceptance condition, this downstream agreement does not define any element of the radiation mapping or transport law.

6.3 Preregistered 23-gate audit

The authority lock records 23 preregistered scientific gates. Table 1 expands the grouped acceptance conditions into their atomic checks and places the frozen threshold beside the corresponding measured or evidence-level result. The labels G01–G23 are manuscript indexing labels only; they do not replace or rename identifiers inside the frozen preregistration or authority archive. For the four species/path runtime checks, the instrumentation counters are path-level aggregates: each instrumented RHS evaluation applies both direct ^{7}Li and ^{7}Be sink terms, and each instrumented analytic-Jacobian evaluation applies the matching diagonal terms for both species. Thus the same nonzero path counter establishes execution of each direct species term on that path.

Table 1: Atomic audit of the 23 preregistered scientific gates. Qualitative freeze/source checks are reported by their preserved evidence decision; numerical checks show the measured residual or counter. Lithium abundance relative to the stellar comparison anchor is not a gate.

Gate	Frozen condition	Threshold / rule	Measured or evidence result	Status
G01	Radiation-response slope	$\lvert dN_{\mathrm{eff}}/d\Delta N_{\mathrm{eff}} - 1\rvert \le 10^{-8}$	6.57×10^{-14}	PASS
G02	Mapping frozen before target rows	Required	Hash-frozen before target execution	PASS
G03	Radiation mapping excludes abundance feedback	Required	Calibration uses reported N_{eff} only	PASS
G04	Gate-off target N_{eff}	absolute error $\le 5 \times 10^{-9}$	4.88×10^{-15}	PASS

Gate	Frozen condition	Threshold / rule	Measured or evidence result	Status
G05	Gate-on target N_{eff}	absolute error $\leq 5 \times 10^{-9}$	4.88×10^{-15}	PASS
G06	Gate-on/off radiation stability	relative change $\leq 10^{-10}$	0 at reported precision	PASS
G07	Internal Y_7 survival	relative error to $1/\pi \leq 0.5\%$	0.0177852%	PASS
G08	Reported mass-seven/H survival	relative error to $1/\pi \leq 0.5\%$	0.0177852% to displayed precision	PASS
G09	D/H gate disturbance	relative change $\leq 10^{-6}$	3.91×10^{-9}	PASS
G10	Y_p gate disturbance	relative change $\leq 10^{-6}$	3.86×10^{-12}	PASS
G11	^{3}He/H gate disturbance	relative change $\leq 10^{-6}$	5.59×10^{-10}	PASS
G12	Gate-off D/H background	$\lvert\text{pull}\rvert \leq 2\sigma$	0.045085σ	PASS
G13	Gate-off Y_p background	$\lvert\text{pull}\rvert \leq 2\sigma$	0.253416σ	PASS
G14	Independent-grid optical depth	relative error $\leq 2 \times 10^{-6}$	1.49682×10^{-7}	PASS
G15	Direct-quadrature normalization	Frozen normalization test	zero reported discrepancy at stored precision	PASS
G16	Live RHS sink, ^{7}Li	nonzero runtime exercise	15,562 aggregate nonzero RHS gate calls	PASS
G17	Live RHS sink, ^{7}Be	nonzero runtime exercise	15,562 aggregate nonzero RHS gate calls	PASS
G18	Analytic-Jacobian sink, ^{7}Li	nonzero runtime exercise	440 aggregate nonzero Jacobian gate calls	PASS
G19	Analytic-Jacobian sink, ^{7}Be	nonzero runtime exercise	440 aggregate nonzero Jacobian gate calls	PASS
G20	Native final-state finiteness	all final states finite	Verified in final audit	PASS
G21	Native final-state nonnegativity	all final states nonnegative	Verified in final audit	PASS
G22	Installed PRIMAT authority source identity	unchanged	Before/after source identity verified unchanged	PASS
G23	Copied patched source import identity	verified copied-source import	Patched copied source verified as imported execution source	PASS

The table is a manuscript expansion of the preregistered decision structure, not a new decision rule. The controlling authority remains the preserved preregistration, frozen mapping, final result, independent audit, and authority lock identified in Section 9.

7 Observed-Coordinate Outcome

The decisive comparison uses PRIMAT-network-native outputs at the jointly transported CPTG baryon and radiation comparison coordinate. Table 2 separates the background-control checks from the report-only lithium diagnostic.

Observable	Comparison value	Native PRIMAT result	Residual	Result
D/H, gate off	$(25.08 \pm 0.29) \times 10^{-6}$	$25.0669253584 \times 10^{-6}$	-0.045085σ	Pass
Y_p^{BBN}, gate off	0.245 ± 0.003	0.245760246933	$+0.253416\sigma$	Pass
^{7}Li/H, gate off	$(1.45 \pm 0.25) \times 10^{-10}$	$5.2735647904 \times 10^{-10}$	$+15.2943\sigma$	Diagnostic
^{7}Li/H, gate on	$(1.45 \pm 0.25) \times 10^{-10}$	$1.6789263563 \times 10^{-10}$	$+0.915705\sigma$	Diagnostic

Table 2: Native PRIMAT outcome at the complete CPTG transported baryon and radiation coordinate. D/H and Y_p are preregistered background gates. Lithium is a downstream diagnostic and is not used to derive or qualify the transport law.

The full chain can be summarized as

$$\begin{gathered}\text{CPTG transported baryon coordinate } + \text{CPTG transported radiation coordinate}\\ \downarrow \\ \text{PRIMAT-network-native gate-off background}\\ \downarrow \\ \text{locked } \tau_7 = \ln\pi \text{ live transport}\\ \downarrow \\ \text{observed lithium comparison band}\\ \text{with negligible disturbance to D/H, } Y_p,\ {}^3\text{He/H, and } N_{\text{eff}}.\end{gathered} \tag{52}$$

No baryon-coordinate retune, radiation-coordinate retune after target exposure, fitted lithium depletion factor, or post-network division by π is used to produce the gated result.

8 Interpretation

The CPTG interpretation is that primordial lithium exposes a selective mass-seven survival channel rather than a failure of the entire light-element abundance system. The transport law acts on the live ^{7}Li and ^{7}Be variables while the network is still evolving. Its integrated strength is geometrically fixed, and the native solver determines the resulting endpoint.

The complete-coordinate PRIMAT result sharpens this interpretation because the transported baryon coordinate, transported radiation coordinate, and mass-seven transport law are all tested together within the network-native solver. The radiation mapping is determined from native N_{eff} calibration alone and frozen before target-row abundance exposure. Its near-zero target residual is therefore a mapping-closure statistic for the frozen implementation, not a new physical prediction of the CPTG radiation coordinate by PRIMAT. At that coordinate, the live gate recovers the locked survival structure while D/H, helium-4, helium-3, and N_{eff} remain effectively unchanged under the transport itself.

The successful result is not accepted because lithium was tuned into agreement. The law is accepted because the fixed $\ln\pi$ optical depth, fixed temporal embedding, prospectively frozen radiation translation, fixed native-network configuration, and frozen numerical gates survive execution before the observational lithium diagnostic is consulted.

9 Evidence and Reproducibility

External scientific sources, the PRIMAT software authority, and observational conventions are cited at their literature or software sources throughout this article. CPTG-generated calculations are associated with named evidence archives, frozen result objects, and checksum identities where an immutable archival release has been established. Repository locations serve as distribution mirrors; for this cryptographically sealed evidence lane, the archive filename together with its SHA-256 digest defines the immutable evidence identity [14].

9.1 PRIMAT Full-Coordinate Primordial-Network Evidence

The accepted complete-coordinate PRIMAT baryon, radiation, and mass-seven validation is bound by the following authority lock.

Authority lock

`CPTG_PRIMAT_A7_FullRadiationCoordinateValidation_AuthorityLock_20260819_2359EDT_r01.zip`

SHA-256

`43b65036e229dc5f04d445f5890399236aaf546c598f1711e572043db6b05bec`

Recognized final-evidence archive

`CPTG_PRIMAT_A7_FullRadiationCoordinateValidation_20260819_2344EDT_r05_FINAL_EVIDENCE.zip`

SHA-256

`9c44543cd2497e72170c0a64173423c9b3bf2f1f3924c9d4e77c16a318fb8b4c`

The recognized final-evidence archive contains the frozen preregistration, the two native radiation-calibration rows, the frozen radiation mapping, the gate-off and gate-on target rows, runtime diagnostics, the final scientific analysis, the independent final audit, and the standard checksum manifest. Its archive manifest verifies 40/40 payload entries with no missing files and no hash mismatches. The authority lock binds this full-coordinate evidence lane to the recognized final-evidence archive and verifies all 23 required scientific gates together with the result-to-mapping and result-to-preregistration bindings.

9.2 Frozen Scientific Identities

The prospective execution is additionally bound to the following frozen scientific objects.

Preregistration SHA-256

`ab38b8c5bdb435b6c311e9b5e0fe31bca1c06ec67d2dc3e4a92098f255425162`

Frozen radiation-mapping SHA-256

`f4e1b0b4e4df47914da9f904d544565ac61a5687879c536560ec8c004b7cbe16`

Final scientific-result SHA-256

`34c0dfdf44d5917ba44d2253c59195dfe258ae4d4de1adf3db6870eafc993626`

Independent final-audit SHA-256

`847f73fe9aa392cfe13ca61135b9570c20b09015caae2ef261de94e8519b7e14`

These identities preserve the prospective ordering of the experiment: preregistration first, abundance-independent radiation mapping second, target execution third, and independent post-execution audit last. They are scientific provenance identifiers; they are not fitting parameters and do not alter the claim boundary stated in this article.

9.3 Evidence Identity and Distribution

For this cryptographically sealed evidence lane, reproducibility is anchored by three elements:

1. the named authority lock and recognized final-evidence archive;
2. their SHA-256 digests together with the frozen scientific-object identities above; and
3. the corresponding calculation, freeze rule, and validation scope stated in this article.

The public CPTG repository and associated distribution locations provide access to the supporting materials [14]. Repository paths are distribution mechanisms rather than scientific identities. The archive filenames and SHA-256 digests above provide the immutable references by which independently obtained copies can be verified byte-for-byte.

10 Validation Scope and Falsifiability

The current result establishes a prospective PRIMAT-network-native validation of the locked CPTG $A = 7$ live transport law at the complete transported BBN comparison coordinate defined by Equations (14), (15), and (16). Its claim boundary is intentionally narrower than a universal statement about every BBN implementation or every cosmological background. In particular, the result validates the network response to the transported CPTG coordinate while retaining PRIMAT's own BBN background expansion.

The mechanism fails under this validation if:

- the PRIMAT radiation translation requires an abundance endpoint or post-target retuning;
- the calibrated radiation response fails the frozen linearity gate or the target N_{eff} fails its absolute tolerance;
- agreement requires replacing $\ln \pi$ by a fitted optical depth;
- either live ^{7}Li or ^{7}Be sink cannot be exercised through the native PRIMAT RHS and analytic Jacobian;

- the measured Y_7 or reported mass-seven/H survival ratio fails the frozen 0.5% target gate;
- D/H, Y_p, or ^{3}He/H changes exceed the frozen 10^{-6} relative control;
- the native gate-off D/H or Y_p background falls outside its frozen 2σ observational gate;
- the independent optical-depth reconstruction fails its frozen tolerance;
- finite/nonnegative native-state, copied-source import, or installed-source identity requirements fail.

The result establishes the joint transported CPTG baryon-plus-radiation coordinate only for the tested PRIMAT v0.3.2 large-network configuration. It does not turn the lithium transport term into a stellar depletion law, a laboratory cross section, or a generic post-processing correction.

The scientific decision is therefore

> **The locked CPTG $A = 7$ live-channel transport law passes prospective PRIMAT-network-native implementation at the complete transported CPTG BBN comparison coordinate, with all 23 preregistered scientific gates satisfied.**

11 Conclusion

The cosmological lithium problem is a selective mass-seven excess: standard primordial networks can preserve deuterium and helium agreement while producing too much surviving mass seven. CPTG addresses that excess with the live-channel transport law fixed by the geometric pi-branch construction [9], rather than by the lithium endpoint itself,

$$Y_{7,\mathrm{CPTG}} = \frac{Y_{7,\mathrm{raw}}}{\pi}, \qquad \tau_{7,\mathrm{trans}} = \ln \pi, \tag{53}$$

implemented dynamically through the ^{7}Li and ^{7}Be equations rather than as a post-processing division.

The primary PRIMAT test is prospective and network-native. The CPTG radiation target is obtained from two native calibration rows without abundance feedback. Before target execution, the mapping is frozen at

$$\Delta N_{\mathrm{eff}}^{\mathrm{CPTG}} = -0.07539322509392317. \tag{54}$$

At the joint transported target $(\Omega_b h^2, N_{\mathrm{eff}}) = (0.021898765370, 2.968584073464)$, PRIMAT reproduces the frozen numerical radiation coordinate to 4.88×10^{-15} absolute and preserves it exactly at the reported precision when the live mass-seven transport is activated.

The resulting mass-seven survival factor is 0.318366498376680, differing from $1/\pi$ by only 0.0177852%. The live gate changes D/H, Y_p, and ^{3}He/H only at the 10^{-9}, 10^{-12}, and 10^{-10} relative levels, respectively. The PRIMAT network lithium prediction moves from

$$(^7\mathrm{Li/H})_{\mathrm{baseline}} = 5.273564790371440 \times 10^{-10} \tag{55}$$

to

$$(^7\mathrm{Li/H})_{\mathrm{gated}} = 1.6789263562731326 \times 10^{-10}, \tag{56}$$

corresponding to $+0.915705\sigma$ under the declared comparison anchor.

All 23 preregistered scientific gates pass. Within the declared test scope, the lithium conclusion is therefore carried by a direct prospective implementation of the complete transported CPTG BBN comparison coordinate together with the locked live mass-seven transport law in the primary nuclear-network authority, while the observational lithium abundance remains strictly downstream of the scientific decision. The present result is a network-native abundance test at transported CPTG coordinates with PRIMAT's native network background retained.

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